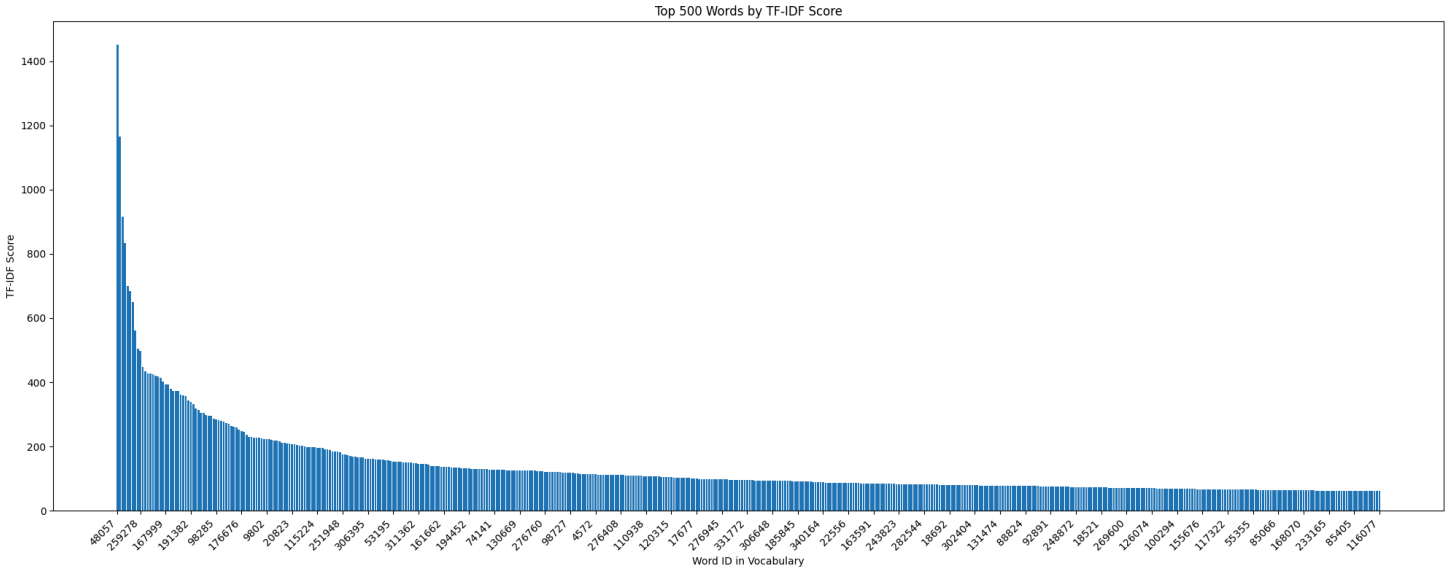
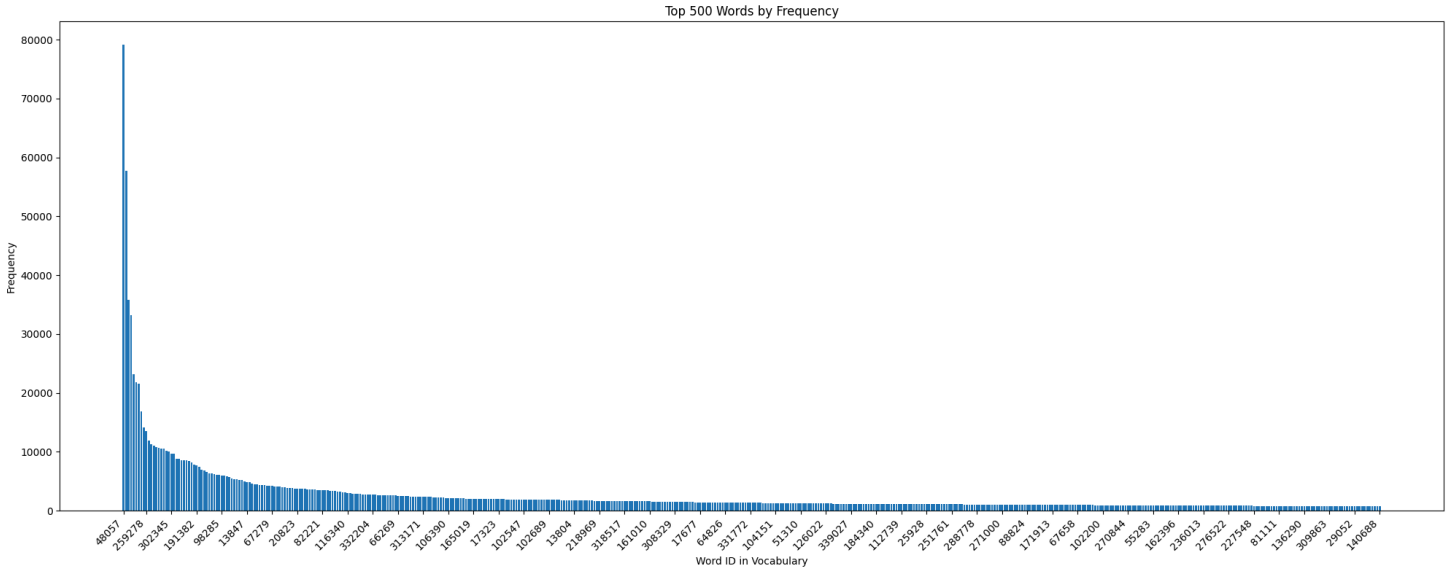
```python
import os
import urllib.request
import jieba
from sklearn.feature_extraction.text import TfidfVectorizer, CountVectorizer
import matplotlib.pyplot as plt
import numpy as np

# 如果在 Colab 環境中，設定支援中文字體避免亂碼
plt.rcParams['axes.unicode_minus'] = False

# Step 1: Download dataset
print("Downloading dataset...")
url = "https://github.com/cjwu/cjwu.github.io/raw/master/courses/nlp/hw1-dataset.txt"
dataset_path = "hw1-dataset.txt"
if not os.path.exists(dataset_path):
    urllib.request.urlretrieve(url, dataset_path)
print("Step 1 complete: Dataset downloaded.")

# Step 2: 讀取語料庫，10行算一個文章，並進行分詞
print("Performing word segmentation...")
documents = []
with open(dataset_path, 'r', encoding='utf-8') as f:
    lines = f.readlines()
    doc_lines = []
    for i, line in enumerate(lines):
        line = line.strip()
        if line:
            doc_lines.append(line)
        # 10行算一個文章
        if (i + 1) % 10 == 0:
            if doc_lines:
                documents.append(" ".join(doc_lines))
                doc_lines = []
    # 處理最後剩餘不足10行的部分
    if doc_lines:
        documents.append(" ".join(doc_lines))

# 進行 Jieba 斷詞
segmented_docs = []
for doc in documents:
    seg_list = jieba.lcut(doc)
    # 過濾空白與換行符號
```

```python
    seg_list = [word for word in seg_list if word.strip()]
    segmented_docs.append(" ".join(seg_list))
print("Step 2 complete: Segmentation done.")

# Step 3: Computing TF-IDF
print("Computing TF-IDF...")
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix = tfidf_vectorizer.fit_transform(segmented_docs)
feature_names = tfidf_vectorizer.get_feature_names_out()
print("Step 3 complete: TF-IDF matrix built.")

# Step 4: Calculating term frequency
print("Calculating term frequency...")
count_vectorizer = CountVectorizer()
tf_matrix = count_vectorizer.fit_transform(segmented_docs)
print("TF calculation done.")

# Step 5: Calculating total TF-IDF and TF scores
print("Calculating total scores...")
tfidf_scores = np.asarray(tfidf_matrix.sum(axis=0)).ravel()
tf_scores = np.asarray(tf_matrix.sum(axis=0)).ravel()

# 建立字詞到編號的對應表 (Word ID in Vocabulary)
word_ids = {word: idx for idx, word in enumerate(feature_names)}

tfidf_dict = {feature_names[i]: tfidf_scores[i] for i in range(len(feature_names))}
tf_dict = {feature_names[i]: tf_scores[i] for i in range(len(feature_names))}
print("Scores calculated.")

# Step 6: 統計前500個高頻和TF-IDF權重高的字詞
top_500_tfidf = sorted(tfidf_dict.items(), key=lambda x: x[1], reverse=True)[:500]
top_500_tf = sorted(tf_dict.items(), key=lambda x: x[1], reverse=True)[:500]

# 定義畫圖函數
def plot_top_words(top_data, title, ylabel, fig_num):
    words = [item[0] for item in top_data]
    y_values = [item[1] for item in top_data]

    # 這裡保留原來的字詞編號 (HW2 規定 x軸: 字詞編號)
    actual_ids = [str(word_ids[w]) for w in words]

    bar_positions = np.arange(len(words))
```

```
    plt.figure(figsize=(20, 8))
    plt.bar(bar_positions, y_values)

    # 抓取 Top 1, 10, 20, 30... 的正確索引 (即 index 0, 9, 19, 29...)
    tick_positions = [0] + list(range(9, len(words), 10))

    #  【嚴格審計修正】: 將原來的「字詞編號」從陣列中抽出來顯示，取代原本寫死的字串
    tick_labels = [actual_ids[i] for i in tick_positions]

    # 將原來的標籤套用至圖表 X 軸
    plt.xticks(tick_positions, tick_labels, rotation=45, fontsize=10, ha='right')
    plt.title(title)
    plt.xlabel("Word ID in Vocabulary")
    plt.ylabel(ylabel)
    plt.tight_layout()
    plt.savefig(f'fig{fig_num}.png')
    plt.show()

# Step 7: Plotting charts
print("Plotting charts...")
# fig#1: TF-IDF權重
plot_top_words(top_500_tfidf, 'Top 500 Words by TF-IDF Score', 'TF-IDF Score', 1)
# fig#2: 出現頻率
plot_top_words(top_500_tf, 'Top 500 Words by Frequency', 'Frequency', 2)
print("Charts saved. Total 2 charts generated.")
```




Top 500 Words by Frequency
Frequency
0
10000
20000
30000
40000
50000
60000
70000
80000
48057
259278
302345
191382
98285
13847
67279
20823
82221
116340
332204
66269
313171
106390
165019
17323
102547
102689
13804
218969
318517
161010
308329
17677
64826
331772
104151
51310
126022
339027
184340
112739
25928
251761
288778
271000
88824
171913
67658
102200
270844
55283
162396
236013
276522
227548
81111
136290
309863
29052
140688
Word ID in Vocabulary